

13. Car Price Prediction Model

AIM

To implement a machine learning model in Python to predict the price of a car based on its features.

SOFTWARE USED

- Python
- Google Colab / Jupyter Notebook
- NumPy
- Pandas
- Matplotlib
- Scikit-learn

PROCEDURE

1. Open Google Colab and create a new notebook.
2. Import the required Python libraries such as Pandas and Scikit-learn.
3. Load the Car Price dataset.
4. Display the dataset and examine its features.
5. Remove unnecessary columns from the dataset.
6. Convert categorical data into numerical values using Label Encoding.
7. Separate the independent variables (X) and dependent variable (y).
8. Split the dataset into training and testing data using `train_test_split()`.
9. Create a Linear Regression model.
10. Train the model using the training dataset.
11. Predict car prices using the testing dataset.
12. Evaluate the model using Mean Squared Error (MSE) and R^2 Score.
13. Display the actual and predicted car prices.

CODE

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score

# Load Car Price dataset

url =
"https://raw.githubusercontent.com/jey1987/DATA605/master/CarPrice_Assignment.csv"

df = pd.read_csv(url)

# Display dataset

print("Dataset Shape:", df.shape)

print("\nFirst 5 rows:")

print(df.head())

# Remove unnecessary columns

df = df.drop(columns=["car_ID", "CarName"])

# Encode categorical columns

encoder = LabelEncoder()

for column in df.select_dtypes(include="object").columns:

    df[column] = encoder.fit_transform(df[column])

# Separate features and target

X = df.drop("price", axis=1)
```

```
y = df["price"]

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create Linear Regression model
model = LinearRegression()

# Train the model
model.fit(X_train, y_train)

# Predict car prices
y_pred = model.predict(X_test)

# Evaluate the model
mse = mean_squared_error(y_test, y_pred)

r2 = r2_score(y_test, y_pred)

print("\nMean Squared Error:", mse)

print("R2 Score:", r2)

# Display actual and predicted price
print("\nActual Price:", y_test.iloc[0])

print("Predicted Price:", y_pred[0])
```

OUTPUT

Dataset Shape: (205, 26)

First 5 rows:

	car_ID	symboling	CarName	fueltype	aspiration	doornumber \
0	1	3	alfa-romero giulia	gas	std	two
1	2	3	alfa-romero stelvio	gas	std	two
2	3	1	alfa-romero Quadrifoglio	gas	std	two
3	4	2	audi 100 ls	gas	std	four
4	5	2	audi 100ls	gas	std	four

	carbody	drivewheel	enginelocation	wheelbase ...	enginesize \
0	convertible	rwd	front	88.6 ...	130
1	convertible	rwd	front	88.6 ...	130
2	hatchback	rwd	front	94.5 ...	152
3	sedan	fwd	front	99.8 ...	109
4	sedan	4wd	front	99.4 ...	136

	fuelsystem	boreratio	stroke	compressionratio	horsepower	peakrpm	citympg \
0	mpfi	3.47	2.68	9.0	111	5000	21
1	mpfi	3.47	2.68	9.0	111	5000	21
2	mpfi	2.68	3.47	9.0	154	5000	19
3	mpfi	3.19	3.40	10.0	102	5500	24
4	mpfi	3.19	3.40	8.0	115	5500	18

	highwaympg	price
0	27	13495.0
1	27	16500.0
2	26	16500.0

3 30 13950.0

4 22 17450.0

[5 rows x 26 columns]

Mean Squared Error: 15916389.725439414

R2 Score: 0.7983838478445078

Actual Price: 30760.0

Predicted Price: 24594.510225239464

The dataset has 205 observations and 26 variables, with price as the dependent variable.

RESULT

Thus, the Car Price Prediction Model was successfully implemented using the Linear Regression algorithm, and car prices were predicted based on the given automobile features.